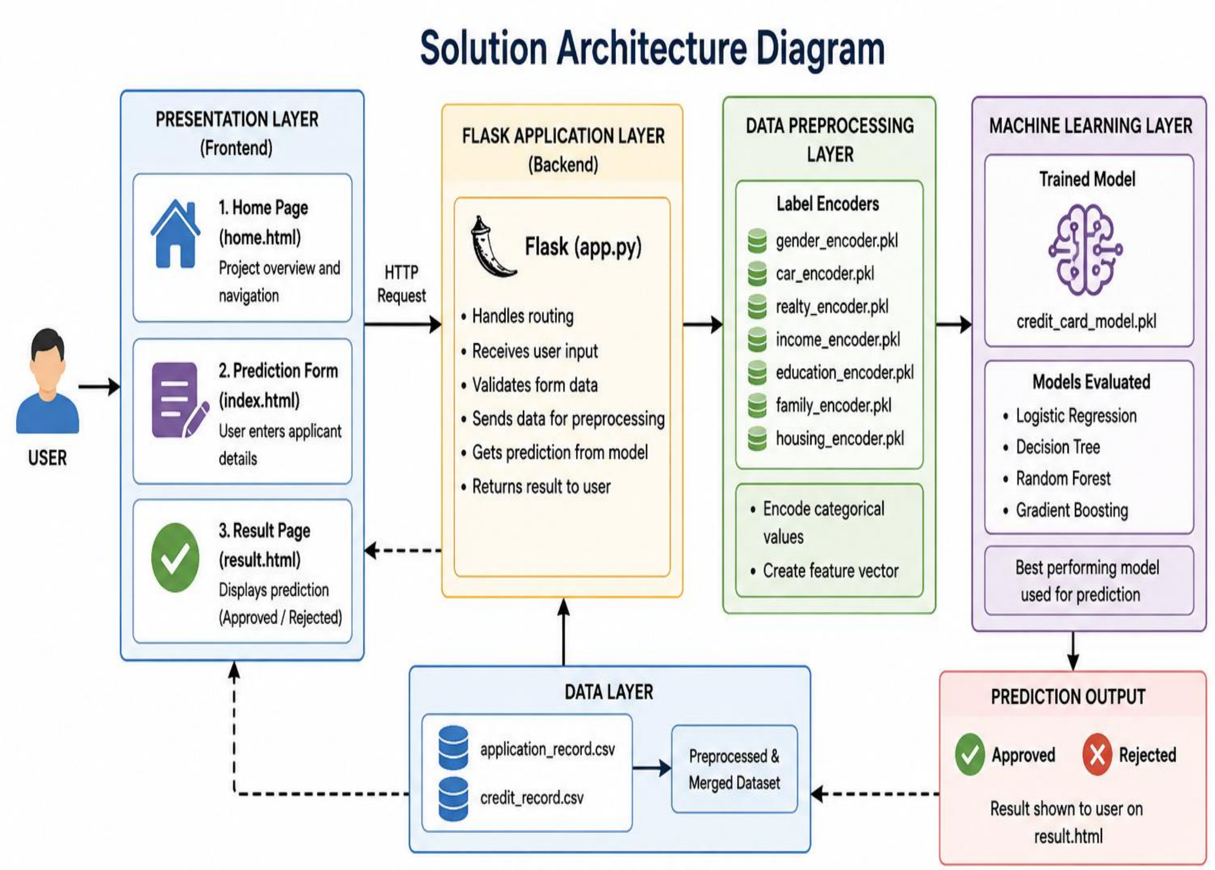
Solution Architecture

Date	28 June, 2026
Team ID	
Project Name	Credit Card Approval prediction
Maximum Marks	5 marks

Solution Architecture Diagram:



Component Description

Component Name	Description / Role	Technologies Used
Presentation Layer	Collects applicant details and displays prediction to the user.	HTML, CSS, Bootstrap, JavaScript
Flask Application Layer	Handles routing, form submission, validation, and communication between frontend and ML model.	Python, Flask
Data Preprocessing Layer	Encodes categorical values using label encoders and prepares feature vector for prediction.	Python, Pandas, NumPy, Joblib
Machine Learning Layer	Loads the trained model and predicts whether the application is approved or rejected.	Scikit-learn, Joblib
Data Layer	Stores original datasets and the processed dataset used for training the model.	CSV Files (Pandas)
Prediction Output	Displays the final prediction result (Approved / Rejected) to the user.	HTML (Jinja2 Templates)

Component Description Table

Component Name	**Description/Role**	**Technologies Used**
Presentation Layer	Collects applicant details and displays prediction.	Flask, HTML, CSS
API Gateway	Handles HTTP requests and routes user input to the prediction logic.	Flask
Core Logic Service	Processes data and predicts approval.	Python, Scikit-learn, Joblib, NumPy
Database	Stores dataset and trained model, and label encoders.	CSV, PKL, Joblib